Arista Access Node Routing Protocol Flap Troubleshooting SOP (WFD-ACC-14)

Node: WFD-ACC-14 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_4714

Description

This Standard Operating Procedure (SOP) addresses the mitigation and resolution of the Routing Protocol Flap failure domain on the Arista Access Node (WFD-ACC-14) located at the Whitefield POP. The failure domain is characterized by BGP route churn, OSPF adjacency loss, and ISIS LSP flooding, as indicated by the ALARM_4714 alarm code with a severity of Warning. The underlying architectural cause of this failure domain lies in the interplay between the routing protocol implementations and the hardware mechanics of the Arista Access Node. Specifically, the BGP route churn may be triggered by the periodic re-advertisement of routes, the OSPF adjacency loss may be caused by the failure to establish or maintain adjacency with neighboring nodes, and the ISIS LSP flooding may result from the excessive generation of Link-State Packets (LSPs) in the presence of network instability. The Arista Access Node, being a critical component of the network infrastructure, plays a pivotal role in the propagation of these routing protocol flaps.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Whitefield POP to provide physical access to the Arista Access Node.

- Verify the status of redundant links to ensure that they are operational and functioning as expected.

- Check the network topology to identify potential bottlenecks or areas of network instability that may be contributing to the routing protocol flap.

Potential Root Causes

Cause	Probability
BGP Route Churn due to Periodic Re-Advertisement	High
OSPF Adjacency Loss due to Failure to Establish/Maintain Adjacency	Medium
ISIS LSP Flooding due to Excessive Generation of LSPs	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version 0, neighbor-count: 2, neighbor-groups: 1, Description: This command isolates the BGP table version, neighbor-count, and neighbor-groups to identify potential issues with BGP route churn.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command identifies OSPF neighbors, their state, and the dead time to diagnose adjacency loss issues.
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```
Command: show isis database, Expected Output: ISIS Level-1 Database, Description: This command isolates the ISIS database to identify excessive generation of LSPs.
```

Resolution Steps

Step 1: Step: Verify BGP Route Churn, Command: show bgp summary, Description: Verify the BGP table version, neighbor-count, and neighbor-groups to identify potential issues with BGP route churn.

Step 2: Step: Adjust BGP Route Churn Thresholds, Command: router bgp 65001 neighbor 10.0.0.1 timers 10 30, Description: Adjust the BGP route churn thresholds to prevent excessive route churn.

Step 3: Step: Repair OSPF Adjacency Loss, Command: router ospf 100 network 10.0.0.0/24 area 0.0.0.0, Description: Repair OSPF adjacency loss by correcting the OSPF network configuration.

Step 4: Step: Mitigate ISIS LSP Flooding, Command: router isis lsp-mtu 1500, Description: Mitigate ISIS LSP flooding by adjusting the LSP MTU.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version 0, neighbor-count: 2, neighbor-groups: 1, Description: Verify that the BGP table version, neighbor-count, and neighbor-groups have been corrected.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: Verify that OSPF neighbors have been restored and are functioning as expected.

Command: show isis database, Expected Output: ISIS Level-1 Database, Description: Verify that the ISIS database has been corrected and LSP flooding has been mitigated.
```

Roll-back Procedures

- Step: Restore Original BGP Route Churn Thresholds, Command: router bgp 65001 neighbor 10.0.0.1 timers 30 60, Description: Restore the original BGP route churn thresholds if the fix fails.

- Step: Restore Original OSPF Network Configuration, Command: router ospf 100 network 10.0.0.0/24 area 0.0.0.0, Description: Restore the original OSPF network configuration if the fix fails.

- Step: Restore Original ISIS LSP MTU, Command: router isis lsp-mtu 1000, Description: Restore the original ISIS LSP MTU if the fix fails.